

Cooling Flows in Galaxy Clusters with a One-Dimensional ZEUS Hydrodynamics Code

Jintong Wang, Jiya Yao

Department of Physics and Astronomy, University of Bologna, Via Gobetti 93/2, 40129, Bologna, Italy

1 Introduction

Galaxy clusters are the largest gravitationally bound structures in the Universe, containing hundreds to thousands of galaxies embedded in a dominant dark matter (DM) halo and a hot, diffuse intracluster medium (ICM). The ICM accounts for a substantial fraction of the baryonic mass in clusters and emits strongly in X-rays due to its high temperatures, typically $10^7 - 10^8$ K (Cimatti et al., 2019). Radiative cooling of this hot plasma can lead to significant energy losses in cluster cores, where the cooling time may become shorter than the Hubble time. In the absence of sufficient heating, the gas is expected to flow inward, forming a classical cooling flow and producing centrally peaked density and X-ray surface brightness profiles.

The evolution of the ICM is governed by hydrodynamical processes driven by gravity, pressure gradients, and radiative cooling. These effects can generate shocks, contact discontinuities, and subsonic inflows, making analytical treatments difficult. Numerical hydrodynamical simulations provide an essential tool for studying cooling flows and the thermodynamic structure of cluster cores. Finite-difference schemes based on operator splitting, such as the ZEUS algorithm (Stone & Norman, 1992), are widely used for astrophysical fluid dynamics because of their robustness and ability to handle compressible flows with strong gradients.

Before applying a numerical solver to cluster scale problems, it is necessary to verify its accuracy using standard test problems with known solutions. A commonly adopted method is the Sod shock tube problem (Stone & Norman, 1992; Hawley et al., 1984; Omang et al., 2006), which evolves an initial discontinuity into a rarefaction wave, a contact discontinuity, and a shock front. Successful reproduction of these features provides a stringent test of the numerical method and ensures reliable treatment of compressible flows and discontinuities.

In this work, we develop a one-dimensional hydrodynamical solver based on the ZEUS algorithm (Stone & Norman, 1992), and implement it in both Cartesian and spherical geometries. The code is first validated using the Sod shock tube and strong shock tests at diverse resolutions. We then apply the validated solver to investigate cooling flows in galaxy clusters, including radiative cooling and gravitational potentials from both a DM halo and a central supermassive black hole (SMBH). The simulations allow us to study the evolution of density, temperature, velocity, X-ray luminosity, and metal enrichment, and to examine the development of a quasi-steady cooling flow in the absence of additional heating processes.

The structure of this paper is as follows. In Section 2 we describe the hydrodynamical equations and numerical method based on the ZEUS algorithm, including the grid setup and boundary conditions, and presents the validation of the code using the Sod shock tube and strong shock tests. In Section 3 we introduce the physical background and setup for the cooling flow simulations, including the gravitational potential, radiative cooling, and initial conditions. Additionally, we present the numerical results of the cooling flow evolution, including density, temperature, velocity, X-ray luminosity, and metal enrichment. Finally, Section 4 summarizes our conclusions.

2 The ZEUS Hydrodynamics Code and the Shock Tube Test

The numerical modeling of astrophysical fluids requires reliable algorithms capable of accurately solving the equations of compressible hydrodynamics. A widely used framework for this purpose is ZEUS-2D, introduced by Stone & Norman (1992). The ZEUS codes solve the Euler equations on an Eulerian grid using a finite-difference method with operator splitting, separating the source and transport terms and updating the physical quantities accordingly. The design of the code emphasizes robustness and flexibility, making it suitable for a wide range of astrophysical fluid problems.

Since this implementation represents a new realization of the ZEUS numerical scheme in one dimension and in different coordinate systems, it is necessary to verify the correctness and performance of the code through a series of hydrodynamical tests. Such tests initially serve to identify possible errors in the implementation. Following this principle, standard test problems are used to validate the hydrodynamical solver. Among them, the Sod shock tube test is one of the most widely adopted benchmarks. The shock tube problem begins with

two regions of gas separated by an initial discontinuity in pressure and density. After the discontinuity is released, the system evolves into a well-defined structure consisting of a shock wave, a contact discontinuity, and a rarefaction wave. Because the analytical solution is known, the test provides a clear and quantitative way to assess whether the numerical scheme can accurately reproduce shock propagation and discontinuities.

In this section, we build a one-dimensional hydrodynamics code inspired by the numerical methods used in ZEUS-2D (Stone & Norman, 1992), and apply the shock tube test to validate the solver in both Cartesian and spherical coordinate systems (Stone & Norman, 1992) with different parameters. This test represents an important step in verifying that the numerical algorithm can correctly capture fundamental hydrodynamical phenomena before it is applied to more complex astrophysical problems.

2.1 The Equations of Motion

To compute astrophysical models, we solve the equations of hydrodynamics, including the continuity equation, the momentum equation, and the internal energy equation.

$$\frac{D\rho}{Dt} + \rho \nabla \cdot \vec{v} = 0, \quad (1)$$

$$\rho \frac{D\vec{v}}{Dt} = -\nabla P - \rho \nabla \Phi - \nabla \cdot \vec{Q}, \quad (2)$$

$$\frac{D\epsilon}{Dt} = -P \nabla \cdot \vec{v} - \vec{Q} : \nabla \vec{v}. \quad (3)$$

Here, the dependent variables are the mass density ρ , the velocity $\vec{v}$, the pressure P , the gravitational potential Φ , and the internal energy density ϵ . The D/Dt denotes the Lagrangian derivative,

$$\frac{D}{Dt} = \frac{\partial}{\partial t} + \vec{v} \cdot \nabla. \quad (4)$$

We adopt an adiabatic equation of state $\epsilon = \frac{P}{\gamma-1}$, where γ is the adiabatic index ($\gamma = C_p/C_v$, where C_p and C_v are the specific heat at constant pressure and volume, respectively). We consider models with different values of γ (see below). The hydrodynamic equations are solved in one-dimensional cartesian coordinates and spherical polar coordinates. To properly capture shock discontinuities in the numerical scheme, an artificial viscosity Q following Von Neumann & Richtmyer (1950) is introduced. Since shock fronts correspond to discontinuities that cannot be resolved on a finite computational grid, the artificial viscosity spreads the shock over a few grid cells, preventing numerical oscillations.

$$Q = \begin{cases} l^2 \rho \left(\frac{\partial v}{\partial x} \right)^2 & \left(\frac{\partial v}{\partial t} < 0 \right), \\ 0 & \text{otherwise,} \end{cases} \quad (5)$$

where l is a constant with dimensions of length which determines the strength of the artificial viscosity. In ZEUS code, the artificial viscosity was written as $Q = C_2 \rho (\Delta v)^2$ (Stone & Norman, 1992).

2.2 Numerical methods

All calculations presented in this work employ the numerical algorithm of ZEUS-2D described by Stone & Norman (1992), adapted here to a one-dimensional case. The hydrodynamical equations are solved on a staggered finite-difference mesh, in which scalar quantities such as the density ρ , internal energy ϵ , and pressure P are defined at cell centers, while the velocity v is defined at cell interfaces. All calculations presented employ the numerical algorithm of ZEUS-2D described by Stone & Norman (1992), adapted here to a one-dimensional case. To sure spatial derivatives and fluxes are evaluated at appropriate locations, improving both the stability and accuracy of the numerical scheme, the hydrodynamical equations are solved on a staggered finite-difference mesh. In this scheme, two spatial grids are introduced. The grid 'a' corresponds to interface with coordinates $x_a(i)$ defined at integer indices $i = 1, 2, \dots, N$, whereas the other grid 'b' corresponds to cell center with coordinates $x_b(i)$ defined at half-integer locations $i - 1/2, i + 1/2, \dots$. The grid spacings are

$$\Delta x_a(i) = x_a(i+1) - x_a(i), \quad \Delta x_b(i) = x_b(i) - x_b(i-1). \quad (6)$$

With this arrangement, physical quantities such as the density ρ , internal energy ϵ , and pressure P are defined at cell centers, while the velocity v is defined at cell interfaces. Besides, in the numerical implementation we adopt the notation $d(i)$, $e(i)$, $p(i)$, $v(i)$ and $q(i)$ to represent density ρ , internal energy ϵ , pressure P , velocity v , and artificial viscosity Q , respectively.

The ZEUS algorithm uses a covariant formulation to allow different coordinate systems to be treated within the same numerical framework (Stone & Norman, 1992). In the general formulation the spatial coordinates are denoted by x_1 , x_2 and x_3 , corresponding to three orthogonal directions of the computational grid. In the present one-dimensional implementation only the coordinate x_1 is retained, while the transverse coordinates x_2 and x_3 are ignored. Consequently, the metric factor g_{32} and the associated transverse volume terms are not required. The geometry enters the equations only through the metric scale factors g_2 and g_{31} and the corresponding volume factors along the x_1 direction.

For Cartesian geometry, where $x_1 = x$, the metric scale factors are constants,

$$g_2(x) = 1, \quad g_{31}(x) = 1. \quad (7)$$

The effective volume factors therefore reduce to the local grid spacing,

$$dvl1a(i) = x_a(i+1) - x_a(i), \quad dvl1b(i) = x_b(i) - x_b(i-1). \quad (8)$$

These quantities represent the widths of the numerical cells centered on the grids 'a' and 'b', respectively.

For spherical symmetry, the coordinate x_1 corresponds to the radial coordinate r . In this case the metric scale factors depend explicitly on position,

$$g_2(r) = r, \quad g_{31}(r) = r. \quad (9)$$

The corresponding volume factors represent the radial shell volumes (up to the constant angular factor 4π , which cancels in the one-dimensional formulation),

$$dvl1a(i) = \frac{x_a(i+1)^3 - x_a(i)^3}{3}, \quad dvl1b(i) = \frac{x_b(i)^3 - x_b(i-1)^3}{3}. \quad (10)$$

With these definitions the same finite-difference scheme can be applied consistently to both Cartesian and spherically symmetric flows. In the following equations the second index used in the original ZEUS-2D formulation is omitted, since only one spatial dimension is considered in the present implementation.

The ZEUS algorithm updates the hydrodynamical variables, in which the evolution over one timestep is divided into a source step and a transport step (Stone & Norman, 1992). The source step accounts for local forces and thermodynamic effects, while the transport step describes the advection of conserved quantities across the grid.

In the present implementation the source step is further decomposed into three substeps. First, the velocity is updated according to the pressure gradient and the gravitational potential (the gravitational term is omitted in the present shock tube tests). The equation (2) can be rewritten as

$$\frac{v_i^{n+a} - v_i^n}{\Delta t} = -\frac{p_i^n - p_{i-1}^n}{\Delta x_{b_i}(d_i^n + d_{i-1}^n)/2}, \quad (11)$$

where $p_i^n = (\gamma - 1)e_i^n$. Second, artificial viscosity effects are considered in regions of compression and used to update both the velocity and the internal energy, allowing shocks to be captured numerically. Thus, equations (2) and (3) are rewritten as

$$\frac{v_i^{n+b} - v_i^{n+a}}{\Delta t} = -\frac{q_i - q_{i-1}}{\Delta x_{b_i}(d_i^n + d_{i-1}^n)/2}, \quad (12)$$

$$\frac{e_i^{n+b} - e_i^n}{\Delta t} = -q_i \frac{v_{i+1} - v_i}{\Delta x_{a_i}}, \quad (13)$$

where n denotes the beginning of the timestep, while $n+a$, $n+b$ represent intermediate states after the pressure, artificial viscosity updates respectively. According to equation (5), the artificial viscosity is

$$q_i = \begin{cases} C_2 d_i (v_{i+1} - v_i)^2 & (v_{i+1} - v_i < 0), \\ 0 & \text{otherwise,} \end{cases} \quad (14)$$

Third, the internal energy is updated to account for compressional heating associated with the velocity divergence. The updated internal energy is

$$e_i^{n+c} = \left[\frac{1 - (\Delta t/2)(\gamma - 1)(\nabla \cdot v)_i}{1 + (\Delta t/2)(\gamma - 1)(\nabla \cdot v)_i} \right] e_i^{n+b}, \quad (15)$$

$$(\nabla \cdot v)_i = \frac{g_{2a_{i+1}} g_{31a_{i+1}} v_{i+1} - g_{2a_i} g_{31a_i} v_i}{dvl1a_i}. \quad (16)$$

The transport step then updates the advection of the hydrodynamical variables across the grid. In the one-dimensional case considered here, only the advection along the x_1 direction is required. A first-order upwind scheme is adopted to evaluate the cell center values according to the sign of the velocity. Here, we should update three variables: density ρ , internal energy ϵ , and momentum ρv . First, the density is updated using the mass flux. The density value at the cell center is chosen using the upwind condition,

$$d_i^* = \begin{cases} d_{i-1} & (v_i > 0), \\ d_i & (v_i \leq 0). \end{cases} \quad (17)$$

The corresponding density flux is

$$F_i^1 = d_i^* v_i g_{2a_i}^{n+1/2} g_{31a_i}^{n+1/2}. \quad (18)$$

The density is then updated as

$$d_i^{n+1} = d_i^n - \Delta t \frac{F_{i+1}^1 - F_i^1}{dv l 1 a_i}. \quad (19)$$

Next, the internal energy is advected. The specific internal energy at the cell center is again determined by the upwind condition,

$$\left(\frac{e}{d}\right)_i^* = \begin{cases} \frac{e_{i-1}}{d_{i-1}} & (v_i > 0), \\ \frac{e_i}{d_i} & (v_i \leq 0). \end{cases} \quad (20)$$

Defining the momentum density, or mass flux, $M_i = d_i^* v_i$, the energy flux becomes

$$F_i^2 = \left(\frac{e}{d}\right)_i^* M_i g_{2a_i}^{n+1/2} g_{31a_i}^{n+1/2}. \quad (21)$$

The internal energy is updated as

$$e_i^{n+1} = e_i^n - \Delta t \frac{F_{i+1}^2 - F_i^2}{dv l 1 a_i}. \quad (22)$$

Finally, the momentum equation is treated in conservative form. The velocity at the cell center is determined using the upwind condition,

$$v_i^* = \begin{cases} v_{i-1} & \left(\frac{v_{i-1} + v_i}{2} > 0\right), \\ v_i & \text{otherwise.} \end{cases} \quad (23)$$

The momentum flux is

$$F_i^3 = v_i^* \frac{M_i + M_{i+1}}{2} g_{2b_i}^{n+1/2} g_{31b_i}^{n+1/2}. \quad (24)$$

The momentum density, defined as $s = \rho v^2$, is then updated as

$$s_i^{n+1} = s_i^n - \Delta t \frac{F_i^3 - F_{i-1}^3}{dv l 1 b_i}, \quad (25)$$

and the velocity is recovered from

$$v_i^{n+1} = \frac{2s_i^{n+1}}{d_i + d_{i-1}}. \quad (26)$$

The second transverse advection step present in the multidimensional ZEUS formulation is not required in the present one-dimensional calculations.

Appropriate boundary conditions are required to ensure numerical stability and a physically consistent evolution of the flow variables. In the present implementation different boundary conditions are applied to vector and scalar quantities. For velocity $\vec{v}$ and momentum $\rho \vec{v}$ defined on grid 'a', reflective boundary conditions are imposed at the inner boundary. In the case of cartesian coordinates, $v_1 = -v_3$ and $v_2 = 0$, where v_2 corresponds to the central velocity, while in spherical geometry, $v_1 = 0$, which enforces zero radial velocity at the origin. At the outer boundary an outflow condition is adopted, $v_N = v_{N-1}$, allowing matter to leave the computational domain without artificial reflections. For scalar quantities such as density ρ and internal energy ϵ , standard outflow boundary conditions are applied at both boundaries $d_1 = d_2$, and $d_N = d_{N-1}$, and similarly for the internal energy e_i .

The timestep is determined dynamically according to a Courant–Friedrichs–Lewy (CFL) stability condition. At each timestep the maximum allowed timestep is computed from the local flow velocity and sound speed,

$$\Delta t = C_{\text{CFL}} \min_i \left(\frac{\Delta x_i}{|v_i| + c_{s,i}} \right), \quad (27)$$

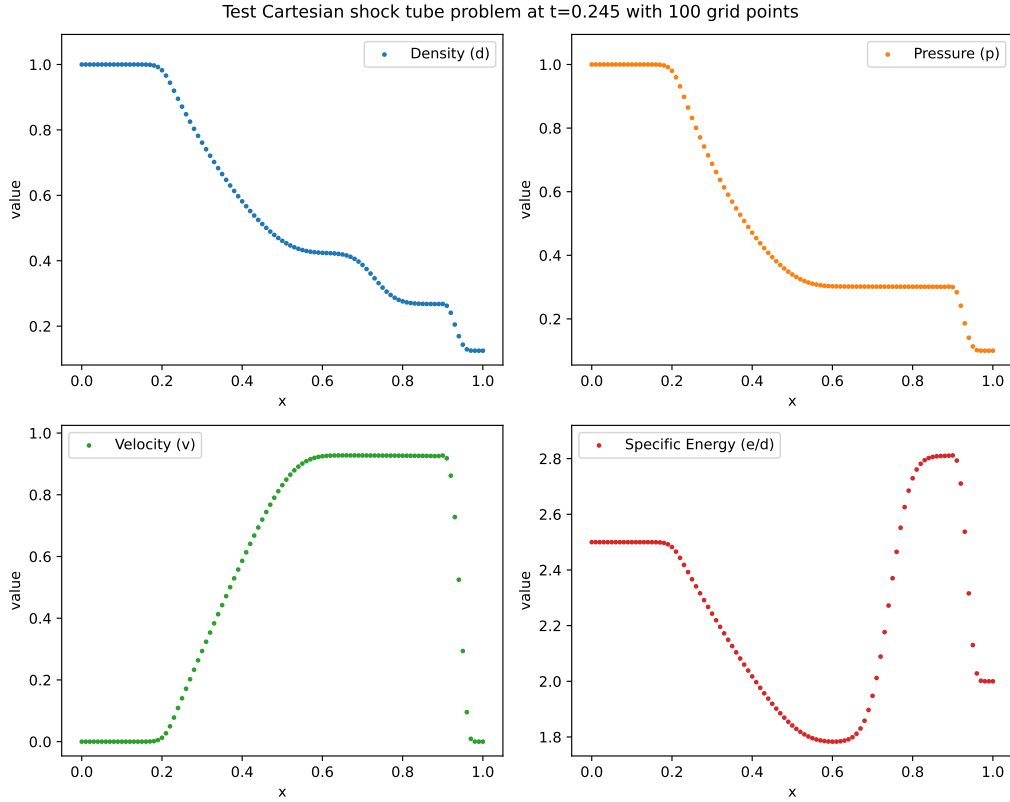


Figure 1: Results for the density (top left), pressure (top right), velocity (bottom left), and specific internal energy e/d (bottom right) profiles for the Sod shock tube problem computed using ZEUS code in one-dimensional cartesian coordinates. A grid of 100 zones is used.

where $c_{s,i} = \sqrt{\gamma P_i / \rho_i}$ is the local sound speed and $C_{\text{CFL}} < 1$ is the Courant number controlling numerical stability. Here, we fixed $C_{\text{CFL}} = 0.5$ in our simulations. This timestep constraint ensures that information does not propagate across more than one computational cell during a single timestep, thereby maintaining the stability and accuracy of the explicit finite-difference scheme.

2.3 The Hydrodynamics Tests

In this subsection, we present the results of the hydrodynamics tests run on the ZEUS code in one-dimensional case. We first present results from simple one-dimensional shock tube problem, involving sod shock tube in cartesian coordinates and spherical shock tube in spherical coordinates. Next we show results of a one-dimensional strong shock tube problem.

2.3.1 Sod Shock Tube Tests

We conduct a one-dimensional problem which tests all the transport and source terms (including artificial viscosity Q). In hydrodynamics, the classical test is the shock-tube problem proposed by Sod (Stone & Norman, 1992). The shock tube problem involves setting up two discontinuous states, a dense gas on the left and a cool rarefied gas on the right, at $i = 0$, and letting them interact. Nonlinear waves are generated at the discontinuity and propagate into each state. A shock wave into the right state and a rarefaction fan into the left. As a test of a numerical algorithm, the shock tube problem demonstrates whether the code can give the correct shock jump conditions and velocity in a stable manner.

The Sod shock-tube problem is a specific set of choices for the initial conditions in the left and right states. For clear visualization, the initial conditions was summarized in Table 1. Initially, the discontinuity is located at $x = 0.5$ at a cell interface. The velocity is initially zero everywhere.

The ZEUS one-dimensional code is applied to the classical Sod shock tube problem. Figures 1 and 2 show the resulting density, pressure, velocity, and specific internal energy profiles obtained using 100 and 1000 computational zones, respectively. The numerical solutions reproduce the main features of the analytic solution (see Fig. 11 of Stone & Norman (1992)), including the rarefaction wave ($x \sim 0.2$), the contact discontinuity ($x \sim 0.7$), and the shock front ($x \sim 0.9$). As expected, increasing the spatial resolution significantly improves

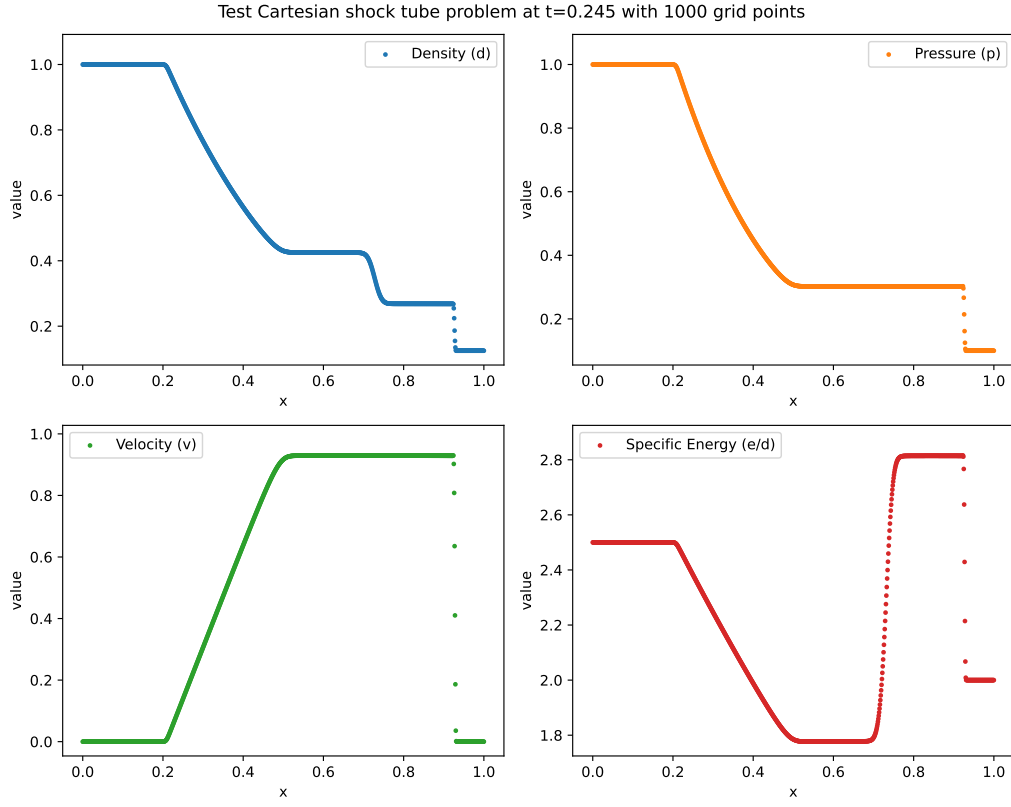


Figure 2: Same as Figure 1, but for 1000 zones.

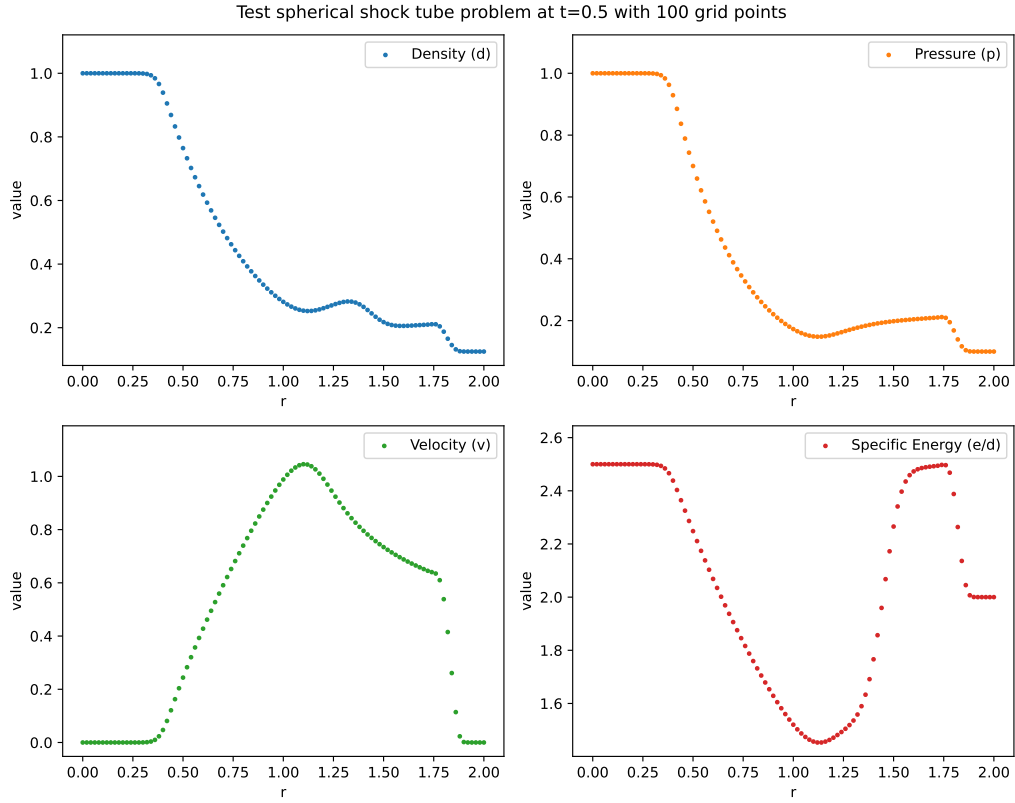


Figure 3: Same as Figure 1, but for spherical coordinates.

the accuracy of the results. In particular, the simulation with 1000 zones provides a much sharper representation of the discontinuities and a closer agreement with the analytic solution compared to the 100 zones case.

Table 1: The initial conditions of shock tube tests.

Tests	γ	ρ_1	P_1	v_1	ρ_2	P_2	v_2	C_2	t_{end}
Sod shock tube (cartesian)	1.4	1.0	1.0	0	0.125	0.1	0	0.3	0.245
Sod shock tube (spherical)	1.4	1.0	1.0	0	0.125	0.1	0	0.3	0.5
Strong shock tube (cartesian)	5/3	100	0.67	0	1.0	0.67×10^{-7}	0	0.3	2.3

Additionally, the ZEUS one-dimensional code is also applied to the Sod shock tube problem in spherical coordinates. A high pressure region is separated from a low pressure region by a spherical membrane located at $r = 1.0$, with the same initial conditions except t_{end} (see Table 1). The results is illustrated in Figures 3 and 4, for the density, pressure, velocity, and specific internal energy in the radial direction using 100 and 1000 zones. These results are in good agreement with the solution of Reimann solver presented in Fig. 3 and 4 of Omang et al. (2006). As expected, increasing the spatial resolution significantly still improves the accuracy of the results.

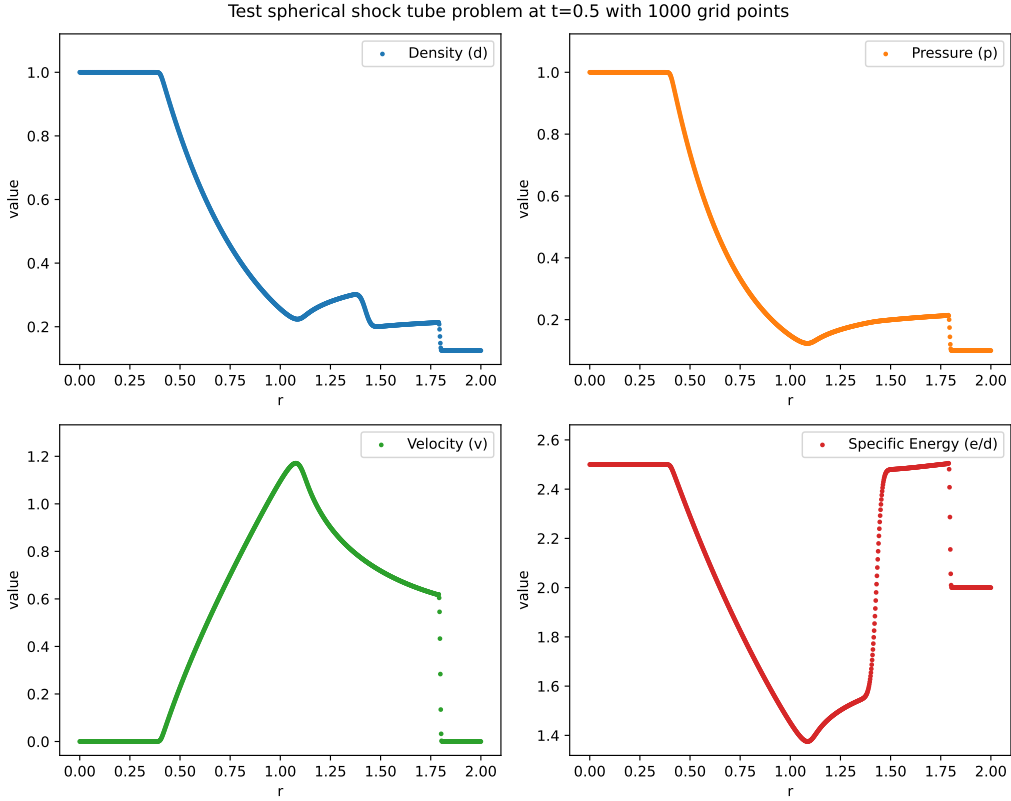


Figure 4: Same as Figure 1, but for spherical coordinates and 1000 zones.

2.3.2 Strong Shock Tube Test

We next conduct the strong shock tube problem in cartesian coordinates using ZEUS code. The discontinuity is set at $x = 0.5$ at a cell interface. Initial conditions are listed in Table 1. This test also applies two spatial resolutions, i.e. 100 zones and 1000 zones. Results are shown in Figures 5 and 6, exhibit diverse behavior. The curves of 100 zones are smoother, while curves of 1000 zones show sharper variation in discontinuities, which is similar to the analytic results shown in fig. 16 and 17 of Hawley et al. (1984).

3 Cooling Flow

3.1 Model

The gas in the cluster is in a state of hydrostatic equilibrium, reflecting a balance between the inward gravitational force and the outward pressure gradient of the gas. In spherical symmetry, the initial gas density profile

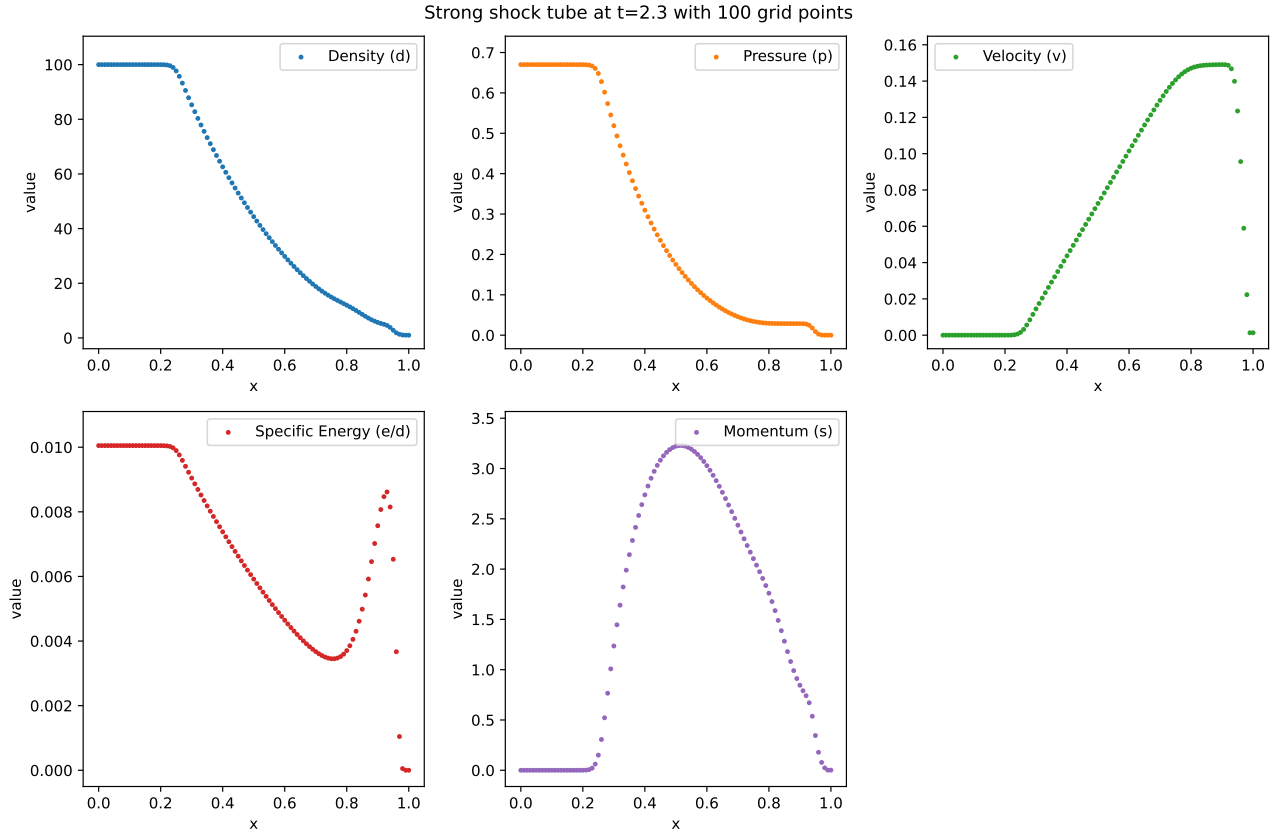


Figure 5: Results of the density (top left), pressure (top middle), velocity (top right), specific internal energy (bottom left), and momentum (bottom middle) profiles for the strong shock tube problem computed using ZEUS code in one-dimensional cartesian coordinates. A grid of 100 zones is used.

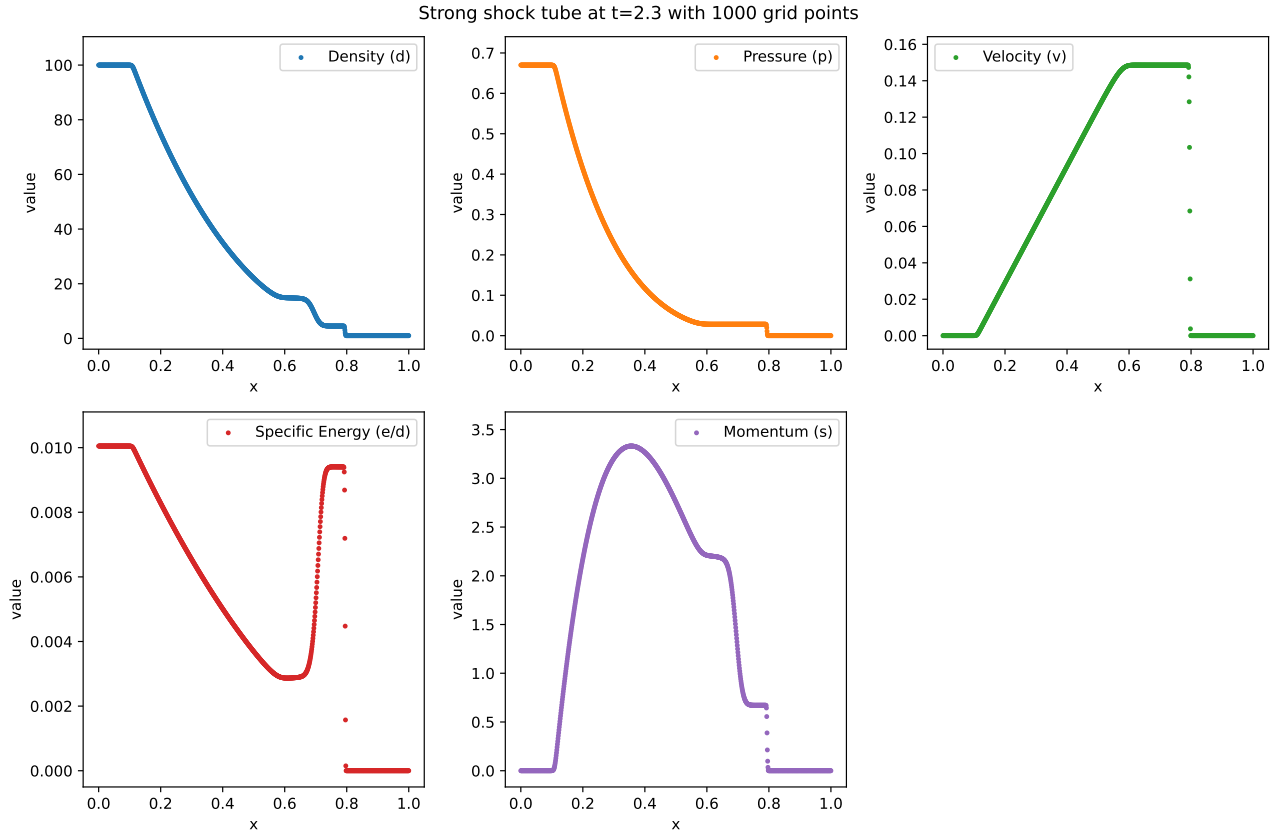


Figure 6: Same as Figure 5, but for 1000 zones.

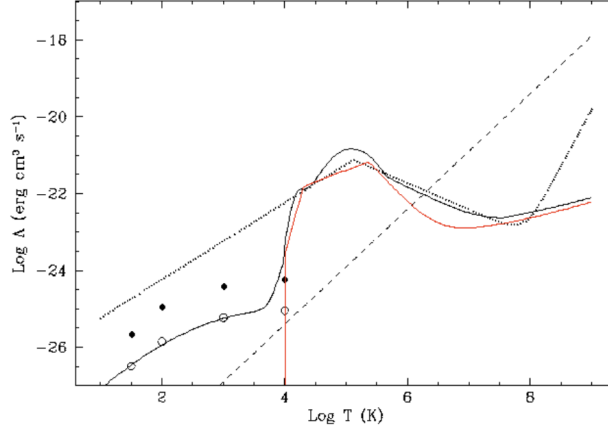


Figure 7: The fitting results of cooling function. The black solid line is used in this work, while the red line is obtained by Sharma et al. (2010).

is obtained by solving the hydrostatic equilibrium equation

$$\frac{dP(r)}{dr} = -\rho_{\text{gas}}(r) \frac{GM(< r)}{r^2}, \quad (28)$$

where P is the gas pressure, $M(< r)$ is the total mass enclosed within a sphere of radius r centered on the cluster, $\rho_{\text{gas}}(r)$ is the gas density, and G is the gravitational constant. Assuming the ICM behaves as an ideal gas,

$$P(r) = \frac{\rho_{\text{gas}}(r) k_B T(r)}{\mu m_p}, \quad (29)$$

where k_B is the Boltzmann constant, $T(r)$ the temperature at r , μ the mean molecular weight, and m_p the proton mass. This guarantees that the ICM is initially in hydrostatic equilibrium and avoids artificial motions at the beginning of the simulation. The details of gas density model were discussed in report I. However, in this work, we additionally considered the gravity provided by the central supermassive black hole (SMBH). The DM NFW model are written as the virial mass form following

$$M_{\text{DM}} = \frac{M_{\text{vir}}(1 - f_b)}{\ln(1 + r_{\text{vir}}/r_s) - \frac{r_{\text{vir}}/r_s}{1 + r_{\text{vir}}/r_s}} \left[\ln(1 + r/r_s) - \frac{r/r_s}{1 + r/r_s} \right], \quad (30)$$

where $M_{\text{vir}} = 1.4 \times 10^{15} M_{\odot}$ denotes the virial mass of the cluster, the baryon fraction is set to $f_b = 0.16296$, consistent with cosmological constraints, $r_{\text{vir}} = 2866.8$ kpc is the virial radius, and $r_s = 449.66$ kpc is the scale radius for NFW halo. The ICM temperature profile is given by

$$\frac{T(r)}{T_{\text{ICM}}} = \frac{1 + (r/71)^3}{2.3 + (r/71)^3} \quad (31)$$

where r is in unit of kpc, and $T_{\text{ICM}} = 8.12 \times 10^7 \text{ K}$.

Radiative energy losses in the hot ICM reduce the thermal pressure of the gas. In the absence of sufficient heating, the gas gradually flows toward the cluster center. This phenomenon is referred to as a cooling flow. It is commonly observed in cool-core clusters, where the gas density is high and the X-ray luminosity peaks in the central region. In real clusters, feedback from the central active galactic nucleus (AGN) can partially offset the cooling.

The radiative energy loss of the gas can be written as (Cimatti et al., 2019)

$$\frac{\partial \varepsilon}{\partial t} = -n_e n_i \Lambda(T, Z), \quad (32)$$

where ε is the internal energy density of the gas, n_e and n_i are the electron and ion number densities, and $\Lambda(T, Z)$ is the cooling function, which depends on the gas temperature T and metallicity Z . In this work, the metallicity in cooling function is assumed to be constant and equal to $0.4 Z_{\odot}$, and the cooling function is the fitting result of the black solid line presented in Figure 7.

The cooling time is the characteristic timescale over which the gas radiates away its thermal energy, which is defined as (Cimatti et al., 2019)

$$t_{\text{cool}} = \frac{3nk_B T}{2n_e n_i \Lambda(T, Z)}, \quad (33)$$

where, n is the total number density. If $t_{\text{cool}} \lesssim$ a few Gyr, the gas can cool efficiently, resulting in condensation of colder gas which potentially enhances star formation or fuels the central AGN.

The dynamics of the ICM are governed by the hierarchy of these timescales. If $t_{\text{cool}} < t_{\text{age}}$, radiative losses become significant preventing the gas from remaining strict hydrostatic equilibrium and indicating a slow inflow toward the cluster center, known as a cooling flow. Here t_{age} is the characteristic age of the system, typically taken as the Hubble time at the observed redshift. The free-fall time, denoted by t_{ff} , represents the characteristic dynamical timescale for gas to fall under gravity from radius r :

$$t_{\text{ff}} \sim \sqrt{\frac{2r}{g(r)}}, \quad (34)$$

where $g(r) = GM/r^2$ is the gravitational acceleration at radius r . When $t_{\text{ff}} < t_{\text{cool}}$, the gas has sufficient time to re-establish pressure equilibrium, remaining approximately hydrostatic. However, in the innermost regions, t_{cool} may drop to the level of the dynamical time ($t_{\text{cool}} \sim t_{\text{ff}}$), triggering a rapid gas inflow. In many observed clusters, this idealized cooling flow is disrupted by mechanical or radiative feedback from the central AGN, which injects energy into the ICM and sustains a state near hydrostatic equilibrium.

A key quantity for characterizing the cooling flow is the mass deposition rate (or cooling rate), denoted as $\dot{M}_{\text{cool}}$, which quantifies the amount of gas transitioning from the hot phase to colder phases per unit time. Assuming the gas cools under isobaric (constant pressure) conditions, the cooling rate can be derived from the X-ray luminosity of the cooling region, since the gas emits strong X-ray L_{cool} at the typical temperatures of the ICM ($10^7 - 10^8$ K) mainly through thermal bremsstrahlung produced by electron-ion interactions,

$$\dot{M}_{\text{cool}}(r) = \frac{2\mu m_p}{5k_B \langle T(r) \rangle} L_{\text{cool}}(r), \quad (35)$$

where μ is the mean molecular weight, m_p is the proton mass, $\langle T(r) \rangle$ denotes the emissivity-weighted temperature profile, which reads

$$\langle T(r) \rangle = \frac{\int_0^r \epsilon_X(r') T(r') dV}{\int_0^r \epsilon_X(r') dV}, \quad (36)$$

where $dV = \frac{4}{3}\pi dr^3$ is the variation of volume. In practice, this analytic estimation serves as a theoretical baseline, which is frequently compared against the effective cooling rates extracted directly from numerical simulations.

To study the cooling rate of the cluster, we can obtain the key information in X-ray band. The classical peaked X-ray surface brightness Σ_X of cooling flows read

$$\Sigma_X(R) = 2 \int_0^R \frac{\epsilon_X(r') r' dr'}{\sqrt{r'^2 - R^2}}, \quad (37)$$

where R is the projected radius along the line of sight, $\epsilon_X(r) = n_e n_i \Lambda(T, r)$ the volume emissivity along the line of sight, 2 means the two side of the cluster. The cumulative X-ray luminosity at radius r can also be derived by

$$L_X(r) = \int_0^r \epsilon_X(r') dV. \quad (38)$$

Hence, the X-ray light curve is also able to derive.

Finally, the iron abundance can also be derived via gas density distribution. The source function $S_{\text{Fe}}(r, t)$ is discussed in report I, and in this work, we consider the time-varying star contribution of iron. The evolution of the iron abundance is solved via hydrodynamics equations utilizing ZEUS code presented in 2. The gas mass is increased by stellar mass return from the BCG, so that the source term accounts for the mass injected into the ICM by stellar winds and explosion of SNIa. The continuity equation (eq. (1)) therefore reads

$$\frac{\partial \rho(r, t)}{\partial t} + \nabla \cdot [\rho(r, t) \vec{v}(r, t)] = \rho_{\star}(r) [\alpha_{\star}(t) + \alpha_{\text{Ia}}(t)]. \quad (39)$$

similarly, the iron density can be written as

$$\frac{\partial \rho_{\text{Fe}}(r, t)}{\partial t} + \nabla \cdot (\rho_{\text{Fe}}(r, t) \vec{v}(r, t)) = \rho_{\star}(r) \left[\frac{\alpha_{\star}(t) Z_{\text{Fe}, \star}(r)}{1.4} + \alpha_{\text{Ia}}(t) \cdot \frac{M_{\text{Fe, Ia}}}{M_{\text{Ia}}} \right]. \quad (40)$$

where ρ_* is the stellar density and α_* and α_{Ia} represent the specific mass return rates from stellar winds and SNIa, respectively. The velocity field of the gas evolves according to the momentum conservation equation (eq. 2). Gas motions arise from the combined effects of pressure gradients, gravitational forces within the cluster potential, and the artificial viscosity. These motions lead to the advection of the gas and energy throughout the computational domain. Gas compression or expansion also contributes to the thermal evolution of the gas through the $P\nabla \cdot \vec{v}$ term in the energy equation (eq. 3), which represents compressional heating or adiabatic cooling. Both stellar mass loss and SNIa explosion inject mass and energy into ICM, contributing to the evolution of the gas energy. The energy equation used in the simulation is

$$\frac{\partial \epsilon}{\partial t} + \nabla \cdot (\epsilon \vec{v}) + P\nabla \cdot \vec{v} = \rho_* \left(e_0 + \frac{1}{2} v^2 \right), \quad (41)$$

where $\rho_* (e_0 + \frac{1}{2} v^2)$ is the stellar source term, representing energy injected by a star and SNIa into the ICM, and the thermal energy e_0 is computed through the average temperature of stellar ejecta, which is determined by $e_0 = C_v T_0$. The average temperature is

$$T_0 = \frac{\alpha_* T_* + \alpha_{\text{SN}} T_{\text{SN}}}{\alpha_* + \alpha_{\text{SN}}}, \quad (42)$$

where α_* and α_{SN} denote the mass return rates from stellar winds and SNIa, and $T_* = \mu m_p \sigma^2 / k_B$ and $T_{\text{SN}} = 1.79 \times 10^9$ K are the star temperature and SNIa temperature, in which $\sigma = 2.5 \times 7$ cm/s is the stellar velocity dispersion.

3.2 Numerical Method

In our simulation, the computational domain is discretized using a one-dimensional radial grid extending from an inner boundary $r_{\text{min}} = 0.1$ kpc to an outer boundary $r_{\text{max}} = 3.5$ Mpc. To capture the high-density gradients in the cluster core while maintaining computational efficiency at large radii, we employ a non-uniform grid of $N_j = 1001$ points. The grid spacing follows a constant expansion factor, $\mathcal{C} = 1.0072$, where the j -th grid interface is defined as:

$$r_j = r_{j-1} + \mathcal{C}(r_{j-1} - r_{j-2}), \quad (43)$$

with an initial spacial resolution of $\Delta r = 50$ pc at the inner boundary.

We adopt a staggered grid consistent with the ZEUS numerical scheme (see section 2.2). Primary physical quantities including gas density, pressure, and temperature are defined at cell centers, $r_{j+1/2}$, while velocities, masses and diffusion fluxes are evaluated at the cell interfaces, r_j . The cell centers are given by $r_{j+1/2} = (r_j + r_{j+1})/2$. Both two set of meshes are utilized throughout the simulation.

Under the assumption of the spherical symmetry, the computational domain is partitioned into a series of concentric shells. The volume of the j -th shell is given by:

$$\Delta V_j = \frac{4}{3} \pi (r_j^3 - r_{j-1}^3). \quad (44)$$

In this work, the gas is subject to gravitational potential of three components: DM halo following a Navarro-Frenk-White (NFW) density profile, the Brightest Cluster Galaxy (BCG), modeled with the Hernquist profile with effective radius $r_{1/2} = 10$ kpc, and a central SMBH with mass $10^9 M_\odot$ treated as a point mass. Details of the BCG model and numerical methods of DM and BCG are presented in report I. The SMBH contributes only via its gravitational potential, and the AGN feedback is not considered in this work.

We solve the hydrodynamic equations using the framework of ZEUS code described in section 2.2, supplemented by source terms of stellar feedback and radiative cooling. Initial density profile is determined by

$$\ln \rho_{\text{gas}, j+1/2} = \ln \rho_{\text{gas}, j-1/2} - (r_{j+1/2} - r_{j-1/2}) \frac{\mu m_p}{k_B (T_{j+1/2} + T_{j-1/2})/2} g_j - (\ln T_{j+1/2} - \ln T_{j-1/2}). \quad (45)$$

In this equation, the temperature entering the hydrostatic equilibrium equation is evaluated at grid interfaces, we utilize the average between adjacent zone-centered points to estimate. Other initial quantities are given by

$$v_j = 0, \quad (46)$$

$$e_{j+1/2} = C_v \rho_{j+1/2} T_{j+1/2}, \quad (47)$$

$$p_{j+1/2} = e_{j+1/2} (\gamma - 1), \quad (48)$$

$$Z_{\text{Fe}, j+1/2} = 0.3 Z_{\text{Fe}, \odot}, \quad (49)$$

$$\rho_{\text{Fe}, j+1/2} = Z_{\text{Fe}, j+1/2} \rho_{j+1/2} / 1.4. \quad (50)$$

Radiative cooling is included in the simulation as a source term in the internal energy equation, evolving with time. Basic numerical discretizations of source term (stellar feedback) is exhibited in report I. Here, we focus on the source step of ZEUS code, which is described in section 2.2. However, we should consider the stellar mass losses ignored by report I. This is the first update of key quantities, following

$$\rho_{j+1/2} = \rho_{j+1/2} + \Delta t \rho_{*,j} (\alpha_* + \alpha_{\text{Ia}}) \quad (51)$$

$$\rho_{\text{Fe},j+1/2} = \rho_{\text{Fe},j+1/2} + \Delta t \rho_{*,j} (\alpha_* Z_{\text{Fe},*}/1.4 + \alpha_{\text{Ia}} Z_{\text{Fe},\text{SN}}) \quad (52)$$

$$e_{j+1/2} = e_{j+1/2} + \Delta t \rho_{*,j} (\alpha_* + \alpha_{\text{Ia}}) (C_v T_0 + \frac{1}{2} v_j^2) \quad (53)$$

where $Z_{\text{Fe},\text{SN}} = M_{\text{Fe},\text{Ia}}/M_{\text{Ia}} = 0.744/1.4$. In the first substep of the source step, we should add the effect of gravity in eq. (11) to update the velocity of gas

$$v_j = v_j - \frac{p_j - p_{j-1}}{\Delta x_{b,j} (d_j + d_{j-1})/2} \Delta t - g_j \Delta t. \quad (54)$$

Once the compressional source step has been completed, the gas temperature T in each cell is recomputed using the ideal-gas relation

$$T_{j+1/2} = \frac{e_{j+1/2}}{C_v \rho_{j+1/2}}. \quad (55)$$

With this updated temperature, we consider the energy that lost via bremsstrahlung, so that the remain energy is

$$e_{j+1/2} = e_{j+1/2} - \Delta t \epsilon_{X,j+1/2}. \quad (56)$$

After updating of energy, we calculate the temperature again via eq. (55), and . Besides, to avoid unphysical overcooling, a temperature floor of 10^4 K is imposed in the simulation. If the calculated temperature falls below this limit, the temperature is reset to 10^4 K. Since this temperature floor artificially modifies the thermodynamic state of the gas, the internal energy must be recomputed to remain consistent with the ideal gas law via eq. (48) to ensure consistency between temperature and internal energy. The pressure is as well as updated via eq. (49).

The transport of gas between neighboring radial cells is computed using a first-order upwind scheme, and the density of gas and iron are updated by this scheme (eq. 17). Next, following the transport step until the energy updated by eq. (22), the temperature should be updated again via eq. (48). Finally, the velocity $v_{j+1/2}$ is obtained by eq.(26).

To compare simulations with observations, we calculate observables involving X-ray projected surface brightness and luminosity. The X-ray projected surface brightness profile is then obtained by summing the contributions of each shell along the line of sight if $T > 10^6$ K,

$$B_{j+1/2} = B_{j-1/2} + 2\epsilon_{X,m+1/2} \left(\sqrt{r_{m+1}^2 - R_{j+1/2}^2} - \sqrt{r_m^2 - R_{j+1/2}^2} \right). \quad (57)$$

The X-ray differential luminosity profile can be derived utilizing the surface brightness, we only consider the contribution of gas with $r > 1$ kpc and $T > 10^6$ K.

$$dL_{X,j+1/2} = dL_{X,j-1/2} + \epsilon_{X,j+1/2} \Delta V_j. \quad (58)$$

Based on the calculation of X-ray luminosity, we are able to derive the cooling rate via the cooling luminosity, with the gas condition $r > 1$ kpc and $T > 10^7$ K. Additionally, the average temperature (eq. (36)) can be calculated by

$$\langle T \rangle_{j+1/2} = \frac{T_{\text{up},j+1/2}}{T_{\text{down},j+1/2}}, \quad (59)$$

where $T_{\text{up},j+1/2} = T_{\text{up},j-1/2} + \epsilon_{X,j+1/2} T_{j+1/2} \Delta V_j$, and $T_{\text{down},j+1/2} = T_{\text{up},j-1/2} + \epsilon_{X,j+1/2} \Delta V_j$. When $\langle T \rangle_{j+1/2} > 0$, the cooling rate is estimated by eq. (35).

For stability, the integration timestep in our simulations is governed by eq. (27). To ensure a smooth transition from the initial hydrostatic state, we employ a time-dependent CFL parameter, $C_{\text{CFL},n+1} = 1.1 C_{\text{CFL},n}$, where $C_{\text{CFL},1} = 0.01$, until it reaches a maximum safety limit of 0.5.

The simulation begins at the cosmic time 8.7 Gyr, which we define as $t = 0$ for the purpose of our evolution analysis. Data outputs are recorded at 1 Gyr intervals to track the long-term evolution of different quantities.

For the cooling flow project, we adopt the central initial gas density $\rho_0 = 0.87 \times 2.17 \times 10^{-24} \text{g cm}^{-3}$. To handle numerical shocks and maintain stability in high-velocity region, we utilize the artificial viscosity with a coefficient of $C_2 = 3$. The chemical evolution is driven by the SNIa rate of $SNu = 0.2$, and the initial metallicity of ICM is set to a sub-solar value of $Z_{\text{Fe,ICM}} = 0.3 Z_{\odot}$, consistent with observed abundance floors in cluster outskirts, and the iron abundance of stellar population is set to $Z_{\text{Fe},*} = 0.8 Z_{\odot}$ (details see report I).

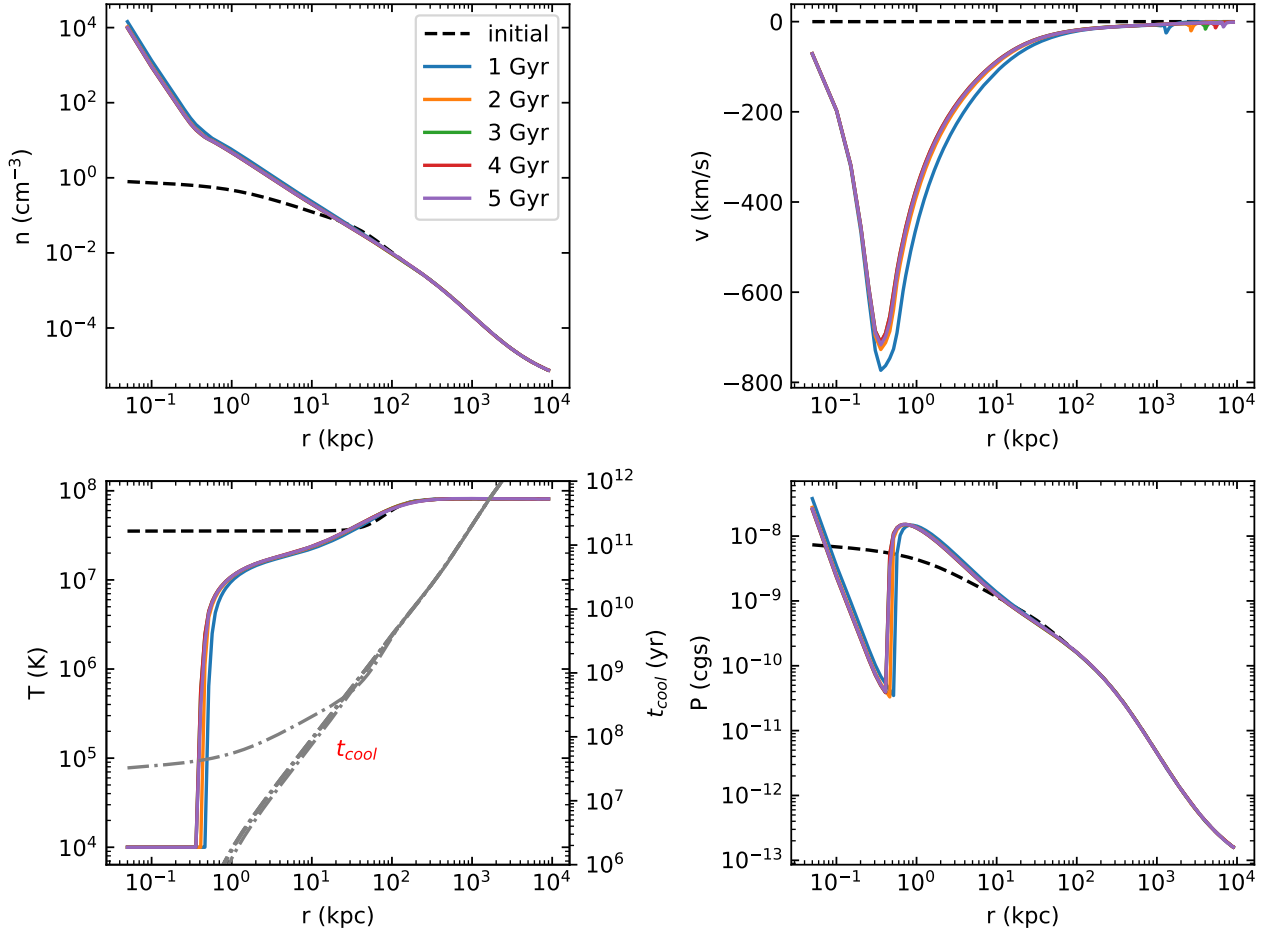


Figure 8: Radial profiles of gas number density (top left), velocity (top right), temperature (bottom left) and pressure (bottom right) from 0 to 5 Gyr with time resolution 1 Gyr. Lines with different colors represent different evolution times. In the bottom left panel (temperature), the gray dash-dotted lines represent the cooling times at various epochs, with the curved one shows t_{cool} at the initial time ($t = 0$).

3.3 Results

3.3.1 Radial Profiles of Global Quantities

The hydrodynamic simulation described in Section 3.1 track the temporal evolution of the gas number density, velocity, temperature, and pressure. While the gas is initialized in hydrostatic equilibrium (see Figure 8), its subsequent evolution is governed by the coupled conservation equations of mass, momentum and energy.

Figure 8 presents the radial profiles of these quantities at 1 Gyr intervals, beginning from the initial time 0 Gyr. The gas density profiles exhibit a monotonic decrease with radii, maintaining a dense core consistent with the initial configuration. The velocity profiles reveal negative radial velocities (inflow) particularly near the center, a signature of classical cooling flows driven by radiative pressure loss. This inward motion is most pronounced near the cooling radius and gradually decelerates as the system evolves. The temperature profiles show a rapid increase in the inner regions before flattening at larger radii, indicating intense radiative cooling where the gas density is highest. We note a near isothermal plateau between 0.1 and 0.4 kpc which is attributed to unphysical reasons. Furthermore, the pressure profile follows the density trend at large radii but exhibits a sharp gradient within the innermost 1 kpc. This localized pressure behavior likely reflects the compression of the inflowing gas effects.

3.3.2 X-ray Surface Brightness and Luminosity

Using eq.(57), we derived the X-ray surface brightness from the simulated density and temperature profiles. As shown in Figure 9, The initial surface brightness follows the gas distribution. Over time, these profiles develop distinct peaks between 0.5 and 0.6 kpc, characteristic of a developing cooling flow. Between 1 and 5 Gyr, the brightness profile remains relatively steady, suggesting the system has reached a quasi-equilibrium state where

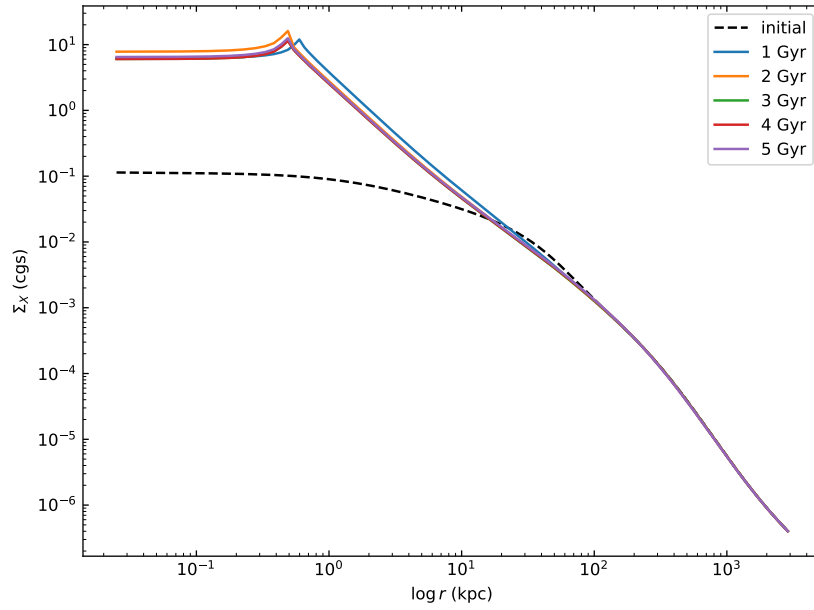


Figure 9: Evolution of the X-ray surface brightness profiles contributed by gas with $T > 10^6$ K for the $\text{SNU} = 0.2$. The dotted line represents the initial state, whereas the solid lines show the results at different times up to 5 Gyr. There is a peak exists at the core of the cluster, indicating features of cooling flow cluster.

radiative losses are balanced by the gravitational compression of the inflowing gas.

We calculate the X-ray luminosity within r_{vir} and $0.5r_{\text{vir}}$. The resulting X-ray light curves are presented in Figure 10). Following an initial transient peak at approximately 0.4 Gyr, the luminosity stabilizes at $\sim 0.6 \times 10^{45}$ erg/s. As seen in the right panel of Figure 10, the zoomed-in light curves exhibit several small peaks rather than a monotonic decrease. Notably, we also observe the slight increasing trend after 3 Gyr.

Finally, the cumulative X-ray luminosity profile is shown in Figure 11. The total X-ray luminosity increases with radius, confirming that the bulk of the X-ray emission originates from the central regions, as indicated by the flattening of the curves at larger radii. The near overlap profiles from 2 Gyr to 5 Gyr suggests that the radiative cooling is effectively balanced by work from the inflow or mechanical feedback, maintaining a steady state in the X-ray regime.

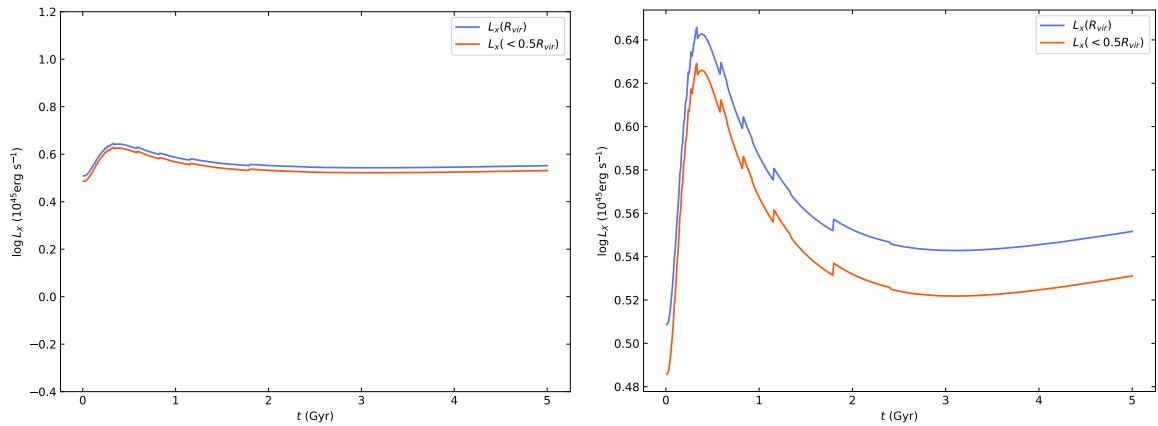


Figure 10: The X-ray light curves originating from gas with $T > 10^6$. The left panel shows the large y-axis scale, while the right panel exhibits the zoomed-in view of the left panel. Blue and red curves represent the integrated luminosity with $r < R_{\text{vir}}$ and $r < 0.5R_{\text{vir}}$, respectively.

3.3.3 Hot and cold mass evolution

To track the thermal evolution, we categorize the gas into two phases based on temperature. Gas with $T < 10^5$ K is defined as cold gas, while hot gas with $T > 10^5$ K. The gas mass in each radial shell is calculated as the

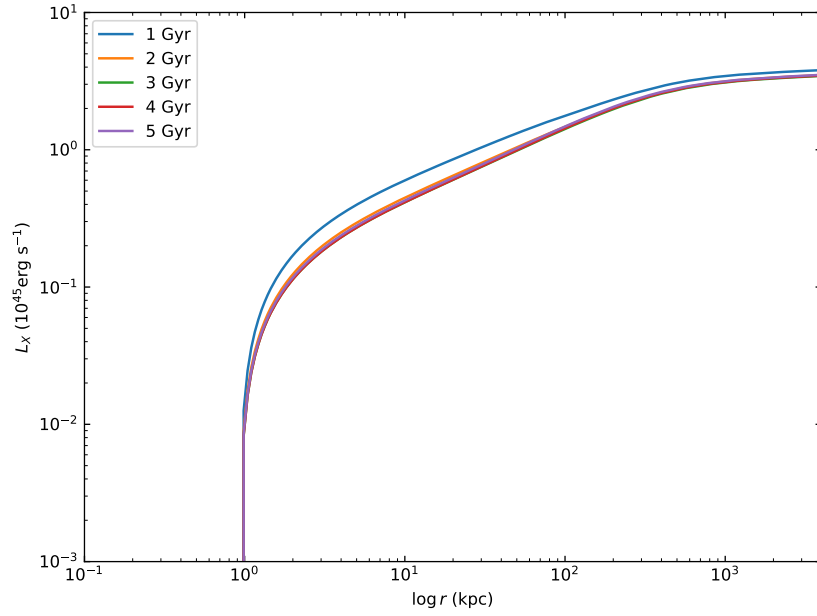


Figure 11: Radial profiles of the cumulative X-ray Luminosity emitted by gas with $T > 10^7$ K. Curves with different colors represent different evolution time. The overlap of profiles from 2 Gyr to 5 Gyr suggests that the system reached a steady state, where radiative cooling is effectively balanced by heating processes such as advection, turbulence, or potential energy release from the inflow.

product of the local gas density and the shell volume. By summing over mass of all shells, the total cold and hot gas masses are obtained. Figure 12 illustrates the time evolution of these mass components. While the hot gas mass remains nearly constant, the cold gas mass increases monotonically. However, the derivative of the cold gas mass gradually decreases over time.

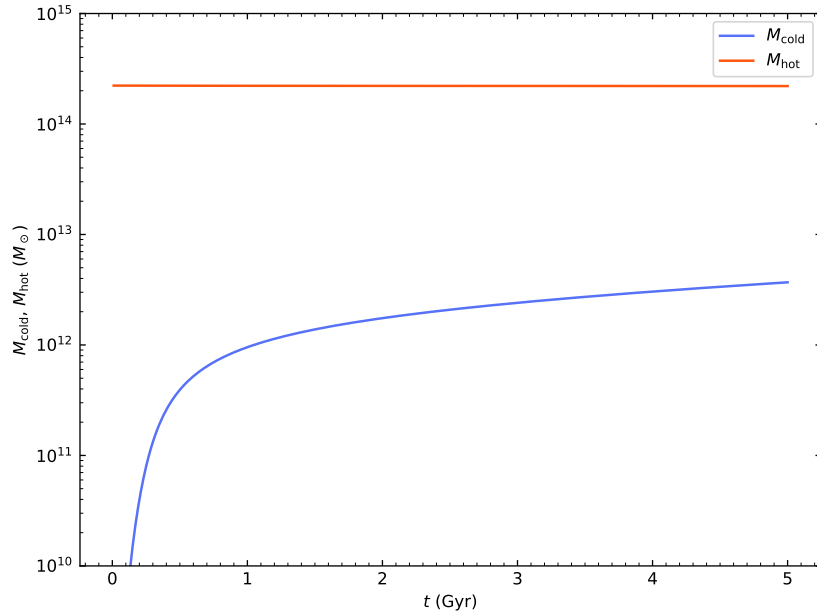


Figure 12: The time series of the integrated gas masses for the hot phase (red solid line) and cold phase (blue solid line). The hot gas mass remains nearly constant, while the cold gas mass increases monotonically due to radiative cooling.

3.3.4 Cooling rate

The simulated cooling rate is derived by

$$\dot{M}_{\text{cool}} = \frac{\Delta M_{\text{cold}}}{\Delta t}. \quad (60)$$

As shown in Figure 13, the cooling rate $\dot{M}_{\text{cool}}(r)$ increases rapidly to a peak at $t \sim 0.4$ Gyr, driven by the high initial central density and short cooling time ($t_{\text{cool}} \propto n^2 \Lambda(T)$). Following this peak, the cooling rate decreases as central gas in the cluster is progressively depleted. Then, the curve becomes relatively flat, indicating the system approaches a quasi-steady state.

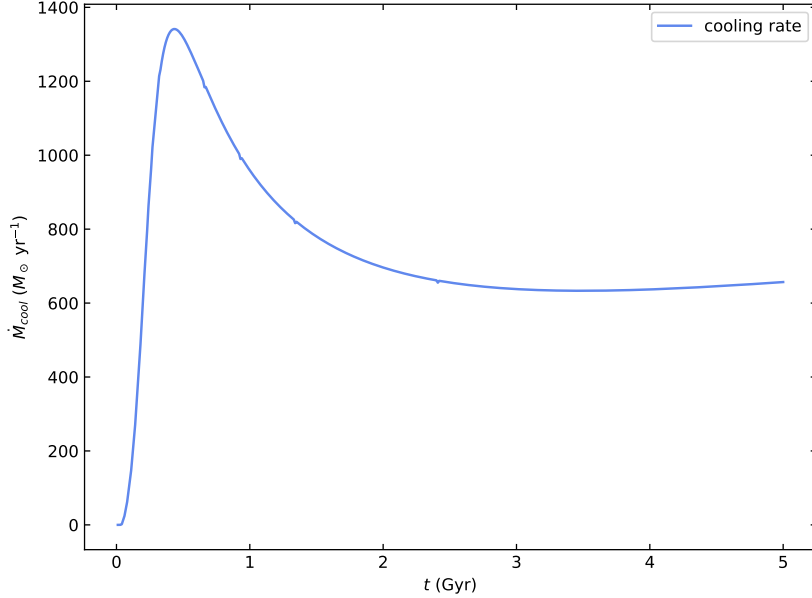


Figure 13: The cooling rate evolution profile. The initial rapid increase and subsequent peak at $t = 0.4$ Gyr are attributed to the high central gas density and short cooling time in the inner region.

For the analytic estimation, the X-ray temperature threshold is set to 10^7 K. We compare the simulated cooling rate to the analytic estimation derived from equation (35). The results of the cooling rate at 5 Gyr in both cases are shown in Figure 14. The purple line represents the cooling rate at 5 Gyr obtained from the analytical formula, while the pink line shows the simultaneous cooling rate derived from simulation. The analytic profile increases with radius primarily. In the innermost region within 1 kpc, the cooling rate is set to be 0 since this region is influenced by AGN while we do not consider its effects. In addition, the analytic cooling rates at different times are shown in Figure 15. The cooling rates from 2 Gyr to 5 Gyr are very similar, which are close to $2700 M_{\odot} \text{ yr}^{-1}$ at the outermost regions.

3.3.5 Fe abundance profile

Figure 16 displays the iron abundance profiles from 1 to 5 Gyr. The left and right panels show the same result but using different scales. From both panels, the overall iron abundance increases over time due to continuous enrichment from the BCG stellar feedback. While the iron is concentrated in the central region, showing a decreasing trend, the profiles exhibit a weak central peak, remaining relatively flat within the range we considered.

4 Conclusion

In this work, we performed one-dimensional hydrodynamical simulations of shock tube test, and a cooling-flow cluster using a ZEUS-type scheme including radiative cooling, stellar mass return, and chemical enrichment. The ICM is initially set in hydrostatic equilibrium within a combined NFW DM halo and central SMBH potential, and subsequently evolves under radiative losses and gravitational compression. Our conclusion are summarized below.

1. We validated the numerical implementation using a standard Sod shock-tube test computed with the one-dimensional ZEUS solver. The simulation successfully reproduces the characteristic structure of the Riemann solution, including the rarefaction wave, contact discontinuity, and shock front. The density, velocity,

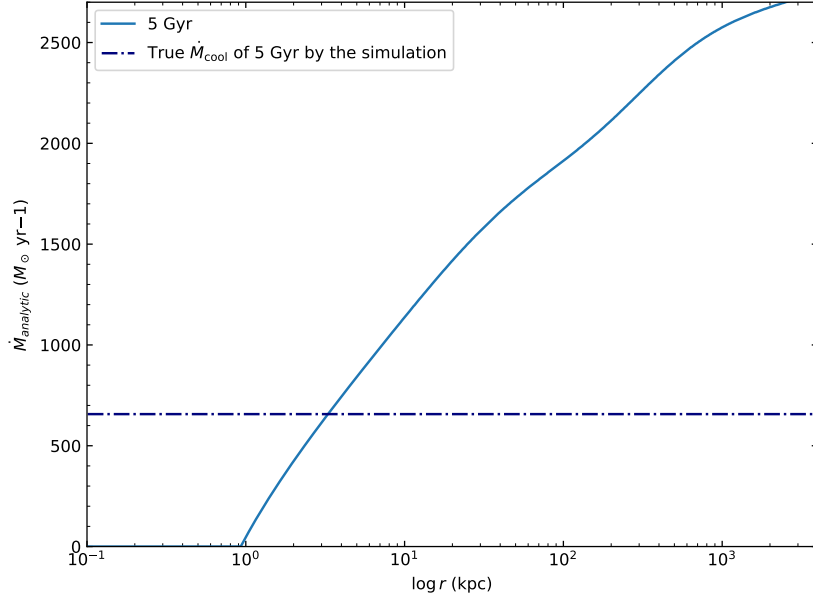


Figure 14: The comparisons of radial cooling rate at 5 Gyr. The purple curve represents the analytical cooling rate obtained from equation (35), while the horizontal pink line shows the simulation result.

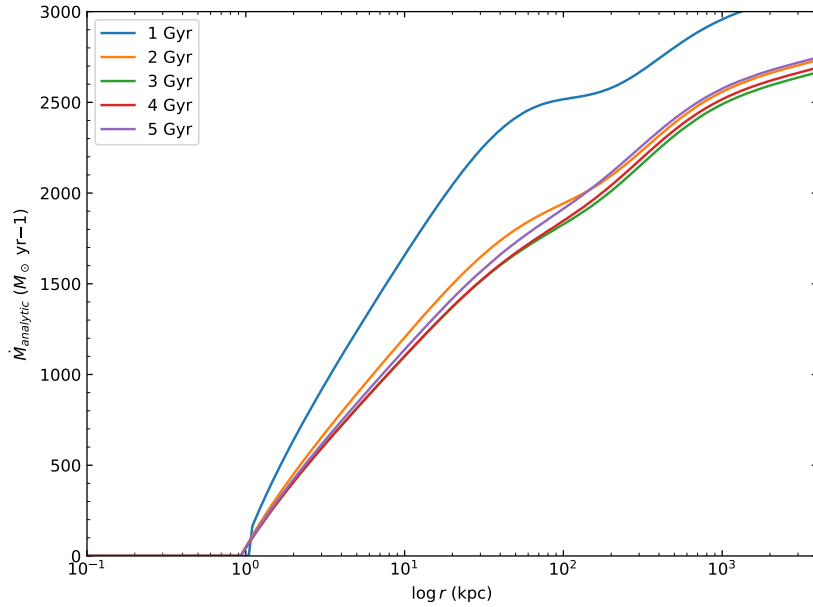


Figure 15: Temporal evolution of the analytical radial cooling rate profiles. Different colors correspond to different epochs, illustrating the gradual approach toward a quasi-steady state.

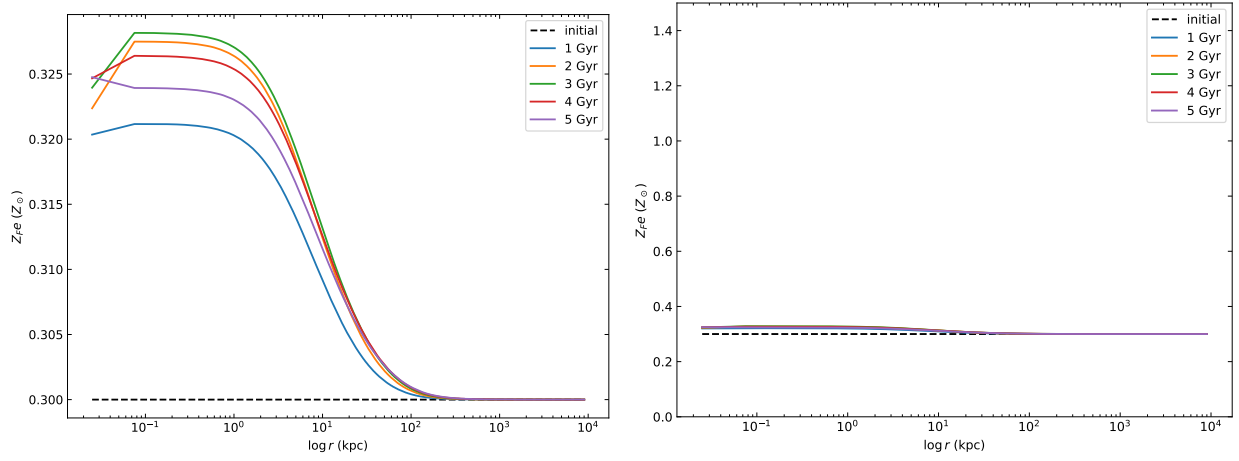


Figure 16: Radial profiles of iron abundance at different evolutionary stages from 0 Gyr to 5 Gyr with $\text{SNU} = 0.2$. The initial abundance profile ($Z_{\text{Fe}} = 0.3Z_{\odot}$) is shown in dotted line. The left and right panels display the same result on different scales.

and pressure profiles show good agreement with the analytic solution, with only mild smoothing across discontinuities due to the artificial viscosity inherent to the ZEUS scheme. The shock position and post-shock states are accurately captured, confirming that the code reliably handles compressible flows with strong gradients. These tests demonstrate that the numerical setup is sufficiently robust for modeling cooling flow dynamics in cluster environments.

2. Radiative cooling drives a gradual inflow toward the cluster center. The radial velocity profiles develop persistent negative values in the inner region, while the density maintains a centrally peaked structure. The temperature profile shows enhanced cooling in the core, accompanied by a steep pressure gradient within the innermost region, consistent with the development of a classical cooling flow.

3. The X-ray surface brightness profiles show a peak in the central region. These profiles are stabilized after the initial transient phase, indicating the system approaches a quasi-steady cooling configuration. The total X-ray luminosity exhibits an early peak followed by a nearly constant level of order $\sim 0.6 \times 10^{45} \text{erg s}^{-1}$, with only mild increase at late times. The cooling rate rises rapidly at early times, reaching a maximum at $t \sim 0.4$ Gyr, and subsequently declines as the central gas reservoir is depleted. At later times, the cooling rate becomes nearly time-independent, indicating the system evolves toward a quasi-steady cooling flow. The analytic cooling rate estimate based on X-ray luminosity agrees broadly with the numerical results and approaches $2700 M_{\odot} \text{yr}^{-1}$ at large radii.

4. Chemical enrichment from stellar mass loss and SNIa gradually increases the iron abundance over time. The resulting profiles show a mild central enhancement with relatively flat radial distribution, consistent with continuous enrichment combined with inward gas transport.

Overall, our simulations reproduce the main characteristics of classical cooling flow clusters: centrally peaked X-ray emission, sustained subsonic inflow, quasi-steady cooling rates, and gradual metallicity buildup. The absence of AGN feedback leads to high cooling rates and long-term accumulation of cooled gas, emphasizing the importance of additional heating processes in realistic cluster cores. Future work including AGN feedback, thermal conduction, and multidimensional effects will be required to obtain a more self-regulated cooling flow solution.

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